

Basic Electronics

Tutorial 1

1. The total charge flowing out of one end of a small copper wire and into an unknown device is determined to follow the relationship $q(t) = 5e^{-t/2}$ C, where t is expressed in seconds. Calculate the current flowing into the device, taking note of the sign. Draw the waveforms of $q(t)$ and $i(t)$ as a function of time.
2. A new type of device appears to accumulate charge according to the expression $q(t) = 10t^2 - 22t$ mC (t in s). (a) In the interval $0 \leq t < 5$ s, at what time does the current flowing into the device equal zero? (b) Sketch $q(t)$ and $i(t)$ over the interval $0 \leq t < 5$ s.
3. The current waveform depicted below Fig. 1 is characterized by a period of 4 s. (a) What is the average value of the current over a single period? (b) Compute the average current over the interval $1 < t < 3$ s. (c) If $q(0) = 1$ C, sketch $q(t)$, $0 < t < 4$ s.

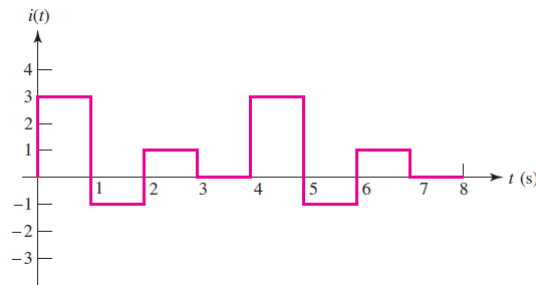


Figure 1: Prob.3

4. Determine the power absorbed by each of the elements in Fig. 2.

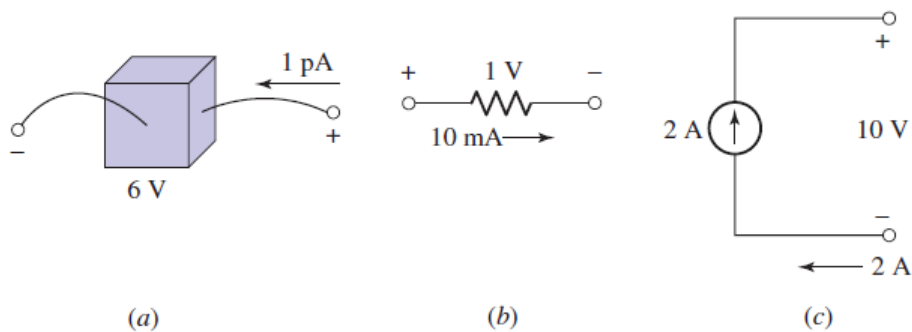


Figure 2: Prob.4

5. Determine the power supplied by the leftmost element in the circuit of Fig. 3.
6. The current-voltage characteristic of a silicon solar cell exposed to direct sunlight at noon in Florida during midsummer is given in Fig. 4. It is obtained by placing different-sized resistors across the two terminals

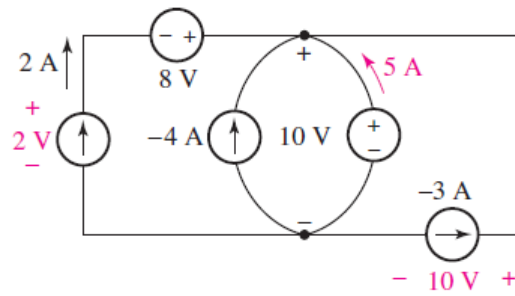


Figure 3: Prob.5

of the device and measuring the resulting currents and voltages. (a) What is the value of the short-circuit current? (b) What is the value of the voltage at open circuit? (c) Estimate the maximum power that can be obtained from the device.

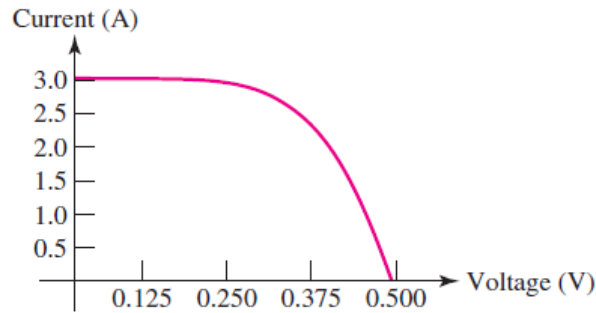


Figure 4: Prob.6

7. Refer to the circuit represented in Fig. 5, while noting that the same current flows through each element. The voltage-controlled dependent source provides a current which is 5 times as large as the voltage V_x . (a) For $V_R = 10$ V and $V_x = 2$ V, determine the power absorbed by each element. (b) Is element A likely a passive or active source? Explain.

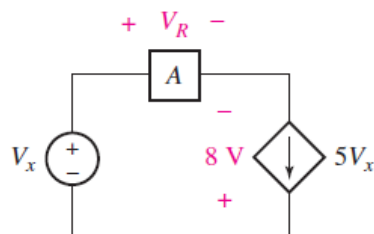


Figure 5: Prob.7

8. The dependent source in the circuit of Fig. 6 provides a voltage whose value depends on the current i_x . What value of i_x is required for the dependent source to be supplying 1 W?

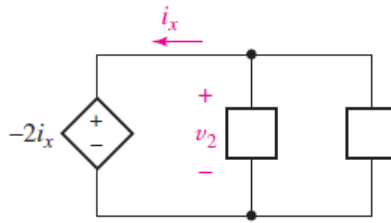


Figure 6: Prob.8

9. Real resistors can only be manufactured to a specific tolerance, so that in effect the value of the resistance is uncertain. For example, a $1\ \Omega$ resistor specified as 5% tolerance could in practice be found to have a value anywhere in the range of 0.95 to $1.05\ \Omega$. Calculate the voltage across a $2.2\ \text{k}\Omega$ 10% tolerance resistor if the current flowing through the element is (a) $1\ \text{mA}$; (b) $4 \sin 44t\ \text{mA}$.
10. For each of the circuits in Fig. 7, find the current I and compute the power absorbed by the resistor.

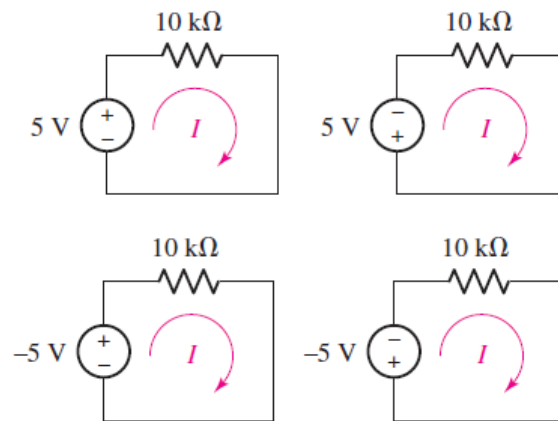


Figure 7: Prob.10

11. Sketch the power absorbed by a $100\ \Omega$ resistor as a function of voltage over the range $-2V \leq V_{\text{resistor}} \leq +2V$.

Basic Electronics

Tutorial 2

- Referring to the circuit depicted in Fig. 1, count the number of (a) nodes; (b) elements; (c) branches.

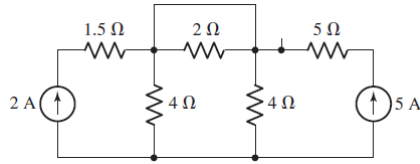


Figure 1: Prob.1

- Refer to the circuit of Fig. 2, and answer the following: (a) How many distinct nodes are contained in the circuit? (b) How many elements are contained in the circuit? (c) How many branches does the circuit have? (d) Determine if each of the following represents a path, a loop, both, or neither: (i) A to B (ii) B to D to C to E (iii) C to E to D to B to A to C (iv) C to D to B to A to C to E

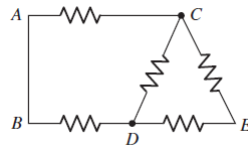


Figure 2: Prob.2

- In the circuit shown in Fig. 3, the resistor values are unknown, but the 2 V source is known to be supplying a current of 7 A to the rest of the circuit. Calculate the current labeled i_2 .

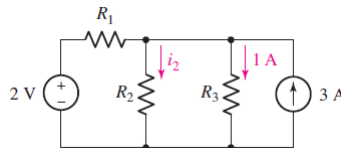


Figure 3: Prob.3

- In the circuit depicted in Fig. 4, i_x is determined to be 1.5 A, and the 9 V source supplies a current of 7.6 A (that is, a current of 7.6 A leaves the positive reference terminal of the 9 V source). Determine the value of resistor R_A .

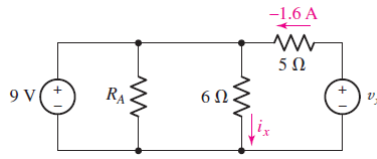


Figure 4: Prob.4

5. Determine the current labeled I_3 in the circuit of Fig. 5.

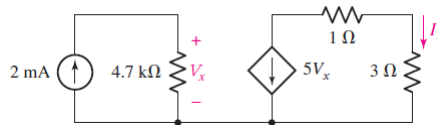


Figure 5: Prob.5

6. In the circuit of Fig. 6, it is determined that $v_1 = 3\text{ V}$ and $v_3 = 1.5\text{ V}$. Calculate v_R and v_2 .

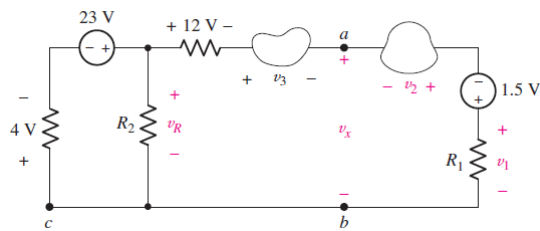


Figure 6: Prob.6

7. (a) Determine a numerical value for each current and voltage (i_1 , v_1 , etc.) in the circuit of Fig. 7. (b) Calculate the power absorbed by each element and verify that they sum to zero.
8. Compute the power absorbed by each element in the circuit of Fig. 8 if the mysterious element X is (a) a $13\ \Omega$ resistor; (b) a dependent voltage source labeled $4v_1$, "+" reference on top; (c) a dependent voltage source labeled $4i_x$, "+" reference on top.

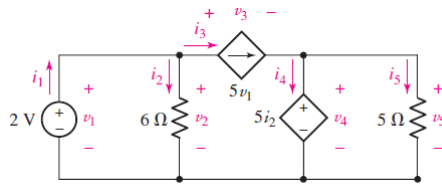


Figure 7: Prob.7

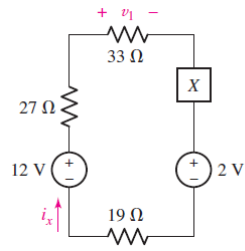


Figure 8: Prob.8

9. Referring to the circuit depicted in Fig. 9 , determine the value of the voltage v .

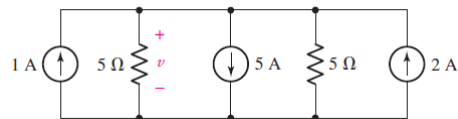


Figure 9: Prob.9

10. Employing resistance combination and current division as appropriate, determine values for i_1 , i_2 , and v_3 in the circuit of Fig. 10.

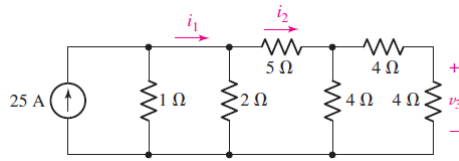


Figure 10: Prob.10

11. In the circuit of Fig. 11, only the voltage v_x is of interest. Simplify the circuit using appropriate resistor combinations and iteratively employ voltage division to determine v_x .

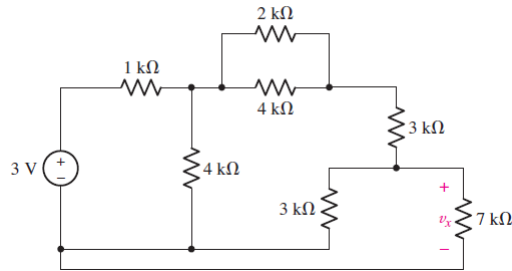


Figure 11: Prob.11

Basic Electronics

Tutorial 3

1. In the circuit of Fig. 1, determine the current labeled i with the assistance of nodal analysis techniques.

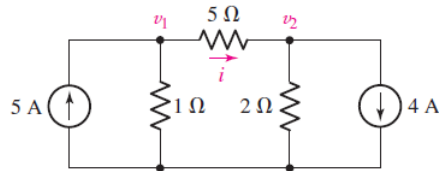


Figure 1: Prob.1

2. Use nodal analysis to find v_P in the circuit shown in Fig. 2

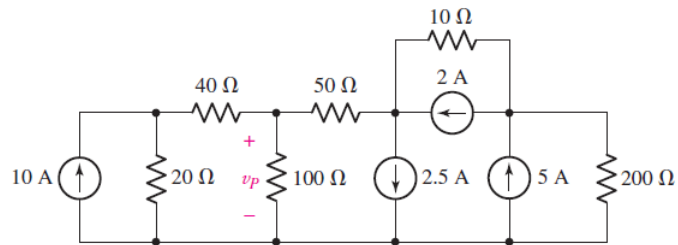


Figure 2: Prob.2

3. For the circuit of Fig. 3, use nodal analysis to determine the current i_5 .

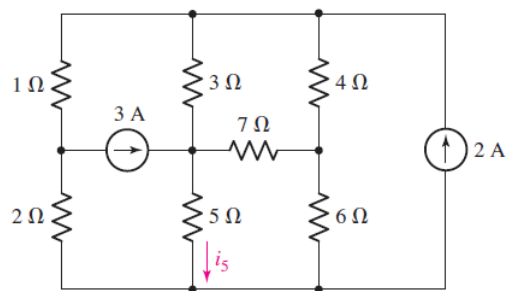


Figure 3: Prob.3

4. Determine the current i_2 as labeled in the circuit of Fig. 4, with the assistance of nodal analysis.

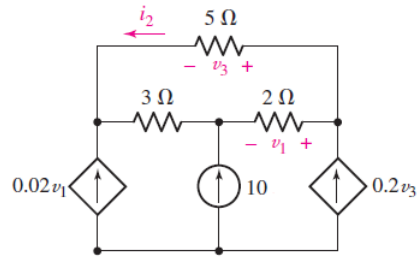


Figure 4: Prob.4

5. Determine the nodal voltages as labeled in Fig. 5, making use of the supernode technique as appropriate.

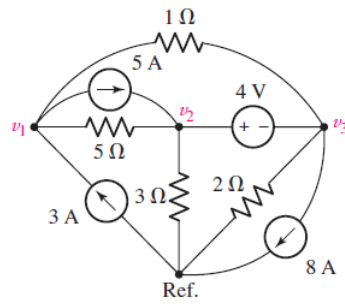


Figure 5: Prob.5

6. Determine the voltage v_x in the circuit of Fig. 6, and the power supplied by the 1 A source.

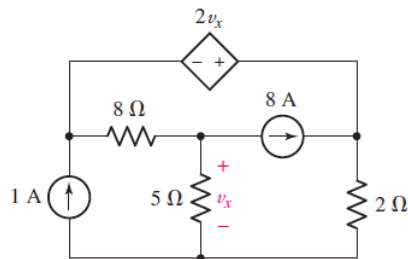


Figure 6: Prob.6

7. For the circuit of Fig. 7, determine all four nodal voltages.

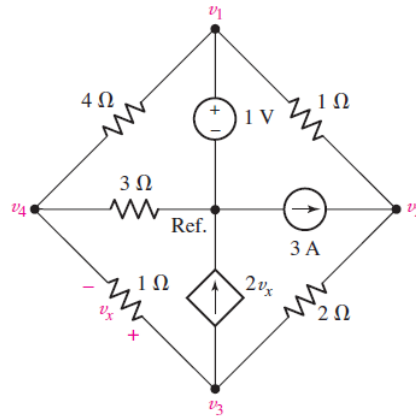


Figure 7: Prob.7

8. Use mesh analysis as appropriate to determine the two mesh currents labeled in Fig. 8

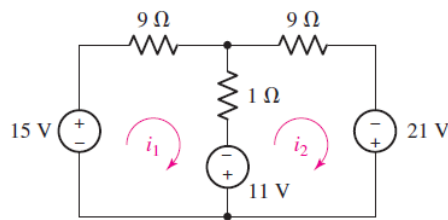


Figure 8: Prob.8

9. Choose nonzero values for the three voltage sources of Fig. 9 so that no current flows through any resistor in the circuit.

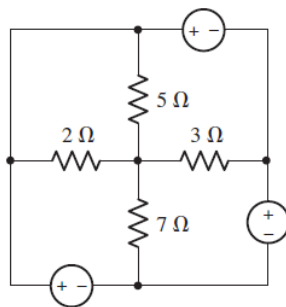


Figure 9: Prob.9

10. Define three clockwise mesh currents for the circuit of Fig. 10 , and employ mesh analysis to obtain a value for each.

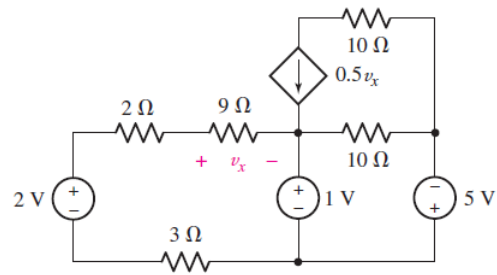


Figure 10: Prob.10

11. Employing the supermesh technique to best advantage, obtain numerical values for each of the mesh currents identified in the circuit depicted in Fig. 11.

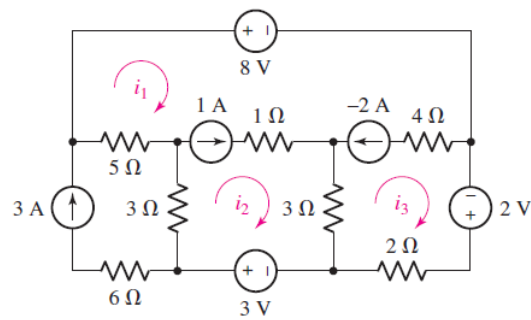


Figure 11: Prob.11

12. Determine the power absorbed by the $10\ \Omega$ resistor in Fig. 12.

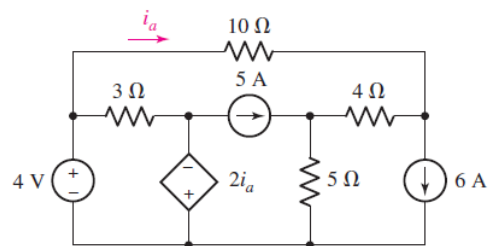


Figure 12: Prob.12

Basic Electronics

Tutorial 4

1. Considering the circuit of Fig. 1, employ superposition to determine the two components of i_8 arising from the action of the two independent sources, respectively.

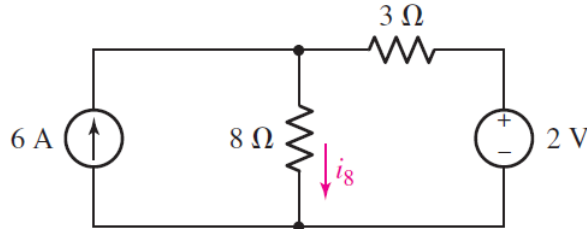


Figure 1: Prob.1

2. (a) Using superposition, determine the voltage labeled v_x in the circuit represented in Fig. 2. (b) To what value should the 2 A source be changed to reduce v_x by 10% ?

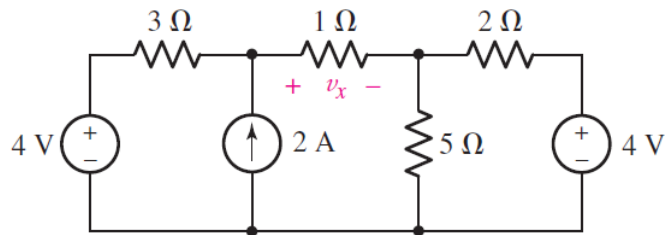


Figure 2: Prob.2

3. (a) Employ superposition to determine the individual contribution from each independent source to the voltage v as labeled in the circuit shown in Fig. 3. (b) Compute the power absorbed by the $2\ \Omega$ resistor.

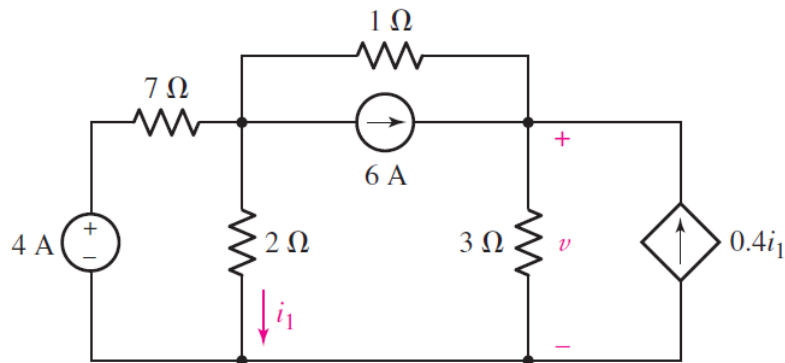


Figure 3: Prob.3

4. For the circuit of Fig. 4, plot i_L versus v_L corresponding to the range of $0 \leq R \leq \infty$.

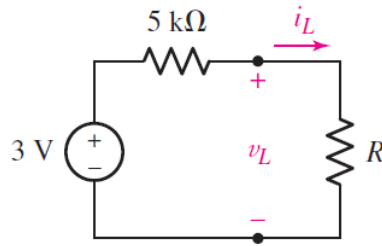


Figure 4: Prob.4

5. Transform the dependent source in Fig. 5 to a voltage source, then calculate V_0 .

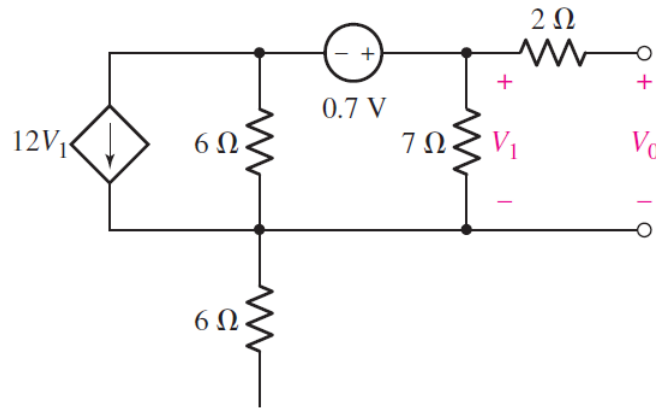


Figure 5: Prob.5

6. With regard to the circuit represented in Fig. 6, first transform both voltage sources to current sources, reduce the number of elements as much as possible, and determine the voltage v_3 .

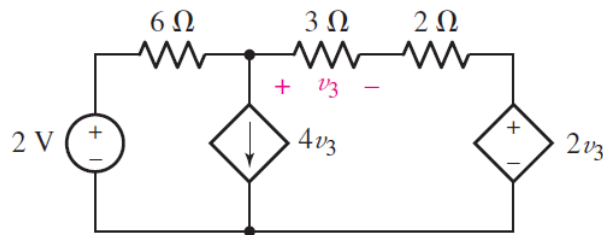


Figure 6: Prob.6

7. Determine the Thévenin equivalent of the network shown in Fig. 7 as seen looking into the two open terminals.
8. (a) Obtain a value for the Thévenin equivalent resistance seen looking into the open terminals of the circuit in Fig. 8 by first finding V_{oc} and I_{sc} . (b) Connect a 1 A test source to the open terminals of the

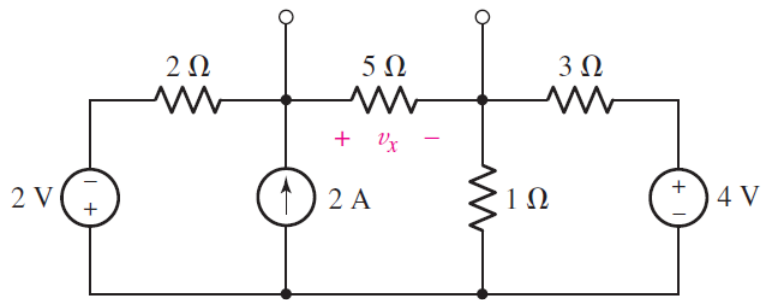


Figure 7: Prob.7

original circuit after shorting the voltage source, and use this to obtain R_{TH} . (c) Connect a 1 V test source to the open terminals of the original circuit after again zeroing the 2 V source, and use this now to obtain R_{TH} .

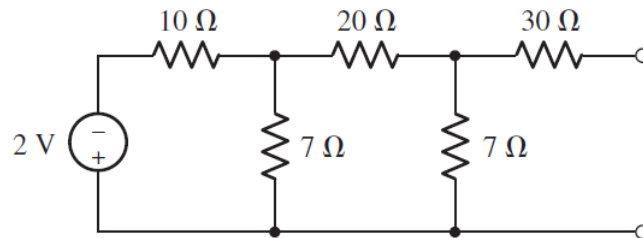


Figure 8: Prob.8

9. Determine the Thévenin and Norton equivalents of the circuit represented in Fig. 9 from the perspective of the open terminals. (There should be no dependent sources in your answer.)

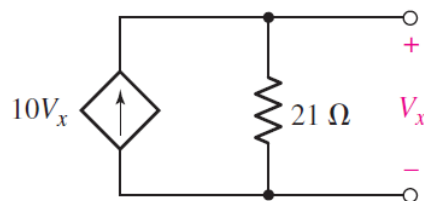


Figure 9: Prob.9

10. Determine the Norton equivalent of the circuit drawn in Fig. 10 as seen by terminals a and b. (There should be no dependent sources in your answer.)

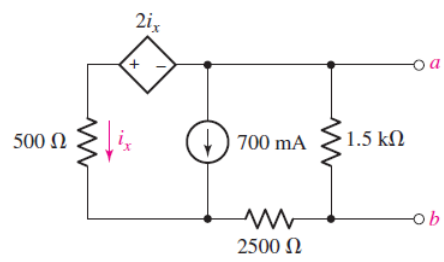


Figure 10: Prob.10

Basic Electronics

Tutorial 5

PN junction diode and applications

Semiconductor Diode

1. a. Determine the thermal voltage for a diode at a temperature of 20°C.
 b. For the same diode of part (a), find the diode current using $I_D = I_s \left(e^{\frac{V_d}{nV_T}} - 1 \right)$ if $I_s = 40 \text{ nA}$, $n = 2$ (low value of V_D), and the applied bias voltage is 0.5 V.
2. a. Using Eq. $I_D = I_s \left(e^{\frac{V_d}{nV_T}} - 1 \right)$, determine the diode current at 20°C for a silicon diode with $n = 2$, $I_s = 0.1 \text{ mA}$ at a reverse-bias potential of -10 V.
 b. Is the result expected? Why?
3. Given a diode current of 6 mA, $V_T = 26 \text{ mV}$, $n = 1$, and $I_s = 1 \text{ nA}$, find the applied voltage V_D .
4. In the reverse-bias region the saturation current of a silicon diode is about $0.1 \text{ } \mu\text{A}$ ($T = 20^\circ\text{C}$). Determine its approximate value if the temperature is increased 40°C .
5. Determine the forward voltage drop across the diode whose characteristics appear in Fig. 1 at temperatures of -75°C , 25°C , 125°C and a current of 10 mA. For each temperature, determine the level of saturation current. Compare the extremes of each and comment on the ratio of the two.

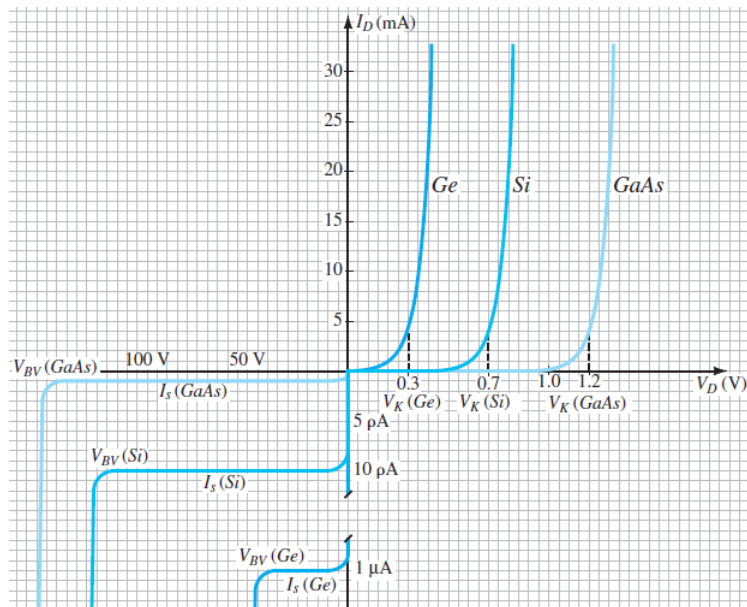


Figure 1:

Resistance Levels

- Determine the static or dc resistance of the commercially available diode of Fig. 2 at a forward currents of 4 mA and 15 mA and then compare results.

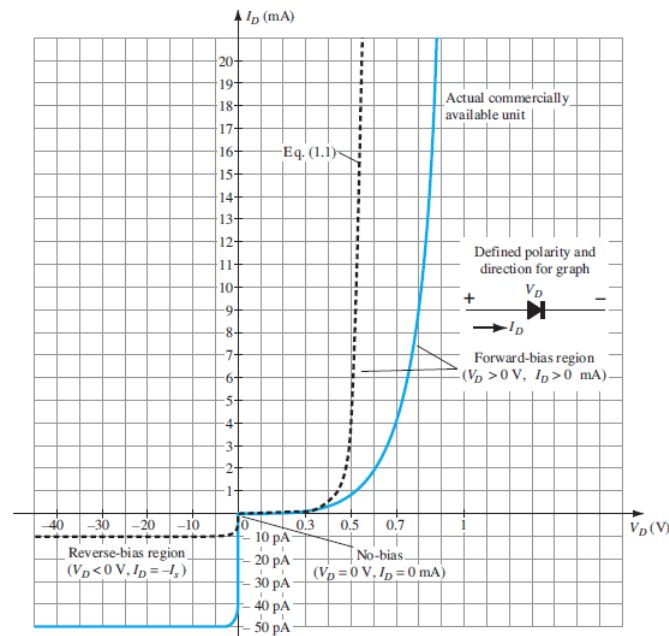


Figure 2:

- Calculate the dc and ac resistances for the diode of Fig. 2 at a forward current of 10 mA and compare their magnitudes.
- Determine the average ac resistance for the diode of Fig. 2 for the region between 0.6 V and 0.9 V. Determine the ac resistance for the same diode of at 0.75 V and compare it to the average ac resistance.

Diode Equivalent Circuits

- Find the piecewise-linear equivalent circuit for the germanium and gallium arsenide diodes of Fig. 1 .

Zener diode

- The following characteristics are specified for a particular Zener diode: $V_Z = 29$ V, $V_R = 16.8$ V, $I_{ZT} = 10$ mA, $I_R = 20$ mA, and $I_{ZM} = 40$ mA. Sketch the characteristic curve.
- Using the curves of Fig. 3 , what level of temperature coefficient would you expect for a 20-V diode? Repeat for a 5-V diode. Assume a linear scale between nominal voltage levels and a current level of 0.1 mA.

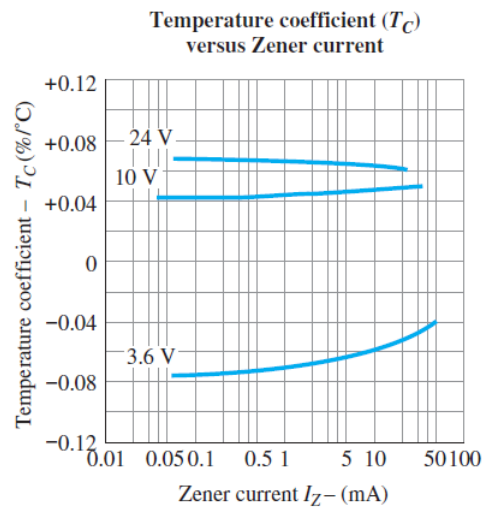


Figure 3:

12. Determine the dynamic impedance for the 24-V diode at $I_Z = 10$ mA for Fig. 4 . Note that it is a log scale.

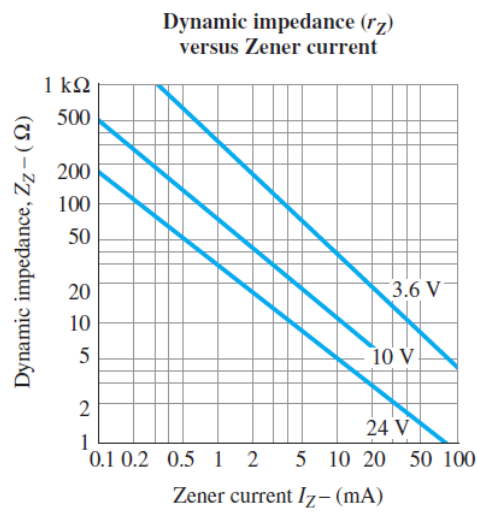


Figure 4: